Fittrack Studio

A Project Report

Submitted in partial fulfilment of the
Requirements for the award of the Degree of

BACHELOR OF SCIENCE (INFORMATION TECHNOLOGY)

By

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Under the esteemed guidance of

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VIVEKANAND EDUCATION SOCIETY'S
COLLEGE OF ARTS, SCIENCE AND COMMERCE
(Autonomous)
Affiliated to University of Mumbai

CHEMBUR MUMBAI – 400071

MAHARASHTRA

2024-2025

Proforma for the Approval Project Proposal

PNR No.: 2022016401914133 Roll No.: 50

1. Name of the Student

Om Bhagwandas Wadhwani

2. Title of the Project

Fittrack Studio

3. Name of the Guide

Mrs. Shital Patil

4. Teaching Experience of the Guide

18 years

5. Is this your first submission? Yes ☐ No ☐

Signature of Student

Date: ……..

Signature of Guide

Date: ……..

Signature of the Coordinator

Date: …….

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COLLEGE OF ARTS, SCIENCE AND COMMERCE
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Affiliated to University of Mumbai

CHEMBUR-MAHARASHTRA-400071

DEPARTMENT OF INFORMATION TECHNOLOGY

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CERTIFICATE

This is to certify that the project entitled, " **Fittrack Studio** ", is bonafide work of **Om Bhagwandas Wadhwani** submitted in partial fulfilment of the requirements for the award of degree of BACHELOR OF SCIENCE in INFORMATION TECHNOLOGY from University of Mumbai.

Internal Guide **Coordinator**

External Examiner

Date: **College Seal**

ABSTRACT

Fittrack Studio is a cutting-edge fitness facility offering a wide range of services, including a fully-equipped gym, specialized boxing and kickboxing programs, personalized training, and a holistic approach to fitness. Before the website's introduction, managing fitness facility operations was inefficient, prone to data security issues, manual errors, delays, and lacked real-time control, underscoring the need for a more streamlined and secure system. The solution is a streamlined, secure, and automated fitness facility management system that enhances data security, minimizes manual errors, provides real-time control, and improves user experience for efficient operations.

This project focuses on developing a streamlined and secure fitness facility management system to address inefficiencies in operations, data handling, and member services. By automating key processes like membership management, payments, the system reduces manual errors and improves overall workflow. It provides role-based access for members, trainers, and administrators, ensuring real-time control and efficient communication. With enhanced data security and a user-friendly interface, the system delivers a seamless experience, promoting better service and operational efficiency.

ACKNOWLEDGEMENT

We thank the people who were a part of this project in numerous ways, people who gave their unending support right from the stage the project idea was conceived.

The four things that go on to make a successful endeavour are dedication, hard work, patience and correct guidance.

We would like to thank our Principal **Dr. Mrs. Anita Kanwar** who has always been the source of inspiration.
We are also thankful to **Mrs. Shital Patil,** our Coordinator who was very much kind enough to give us an idea and guide us throughout our project work.

We take this opportunity to offer sincere thanks to **Mrs. Shital Patil.**

We are thankful to all the teaching staff (I.T) who shared their experience and gave their suggestions for developing our project in a better way.

We are also thankful to **Mrs. Divya Shetty** for helping us out in Project Documentation.

DECLARATION

I hereby declare that the project entitled, " **Fittrack Studio** " has not been in any case duplicated or submitted to any other university for the award of any degree. To the best of my knowledge, no one other than me has submitted this project to any other university. The project is done in partial fulfilment of the requirements for the award of the degree of BACHELOR OF SCIENCE (INFORMATION TECHNOLOGY) and is submitted as the final semester project as part of our curriculum.

Name and Signature of the Student

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CHAPTER 1

INTRODUCTION

1.1 Background:

Fittrack Studio is a state-of-the-art fitness facility designed to cater to a wide range of fitness enthusiasts, from beginners to seasoned athletes. The studio offers a comprehensive suite of services including a fully-equipped gym, specialized boxing training, and dynamic kickboxing sessions. Fittrack Studio aims to create a holistic fitness environment that promotes physical health, mental well-being, and a strong sense of community.

The gym at Fittrack Studio is outfitted with the latest fitness equipment, providing members with everything they need for strength training, cardio workouts, and functional fitness. Certified personal trainers are available to offer personalized workout plans and guidance, ensuring that each member can achieve their individual fitness goals.

In addition to the gym, Fittrack Studio boasts a robust boxing program. This program is designed to accommodate everyone from novices to experienced boxers. Professional coaches lead training sessions that focus on improving technique, agility, and endurance. Regular sparring sessions and boxing fitness classes offer members a chance to practice their skills and enhance their cardiovascular health.

The kickboxing program at Fittrack Studio combines martial arts techniques with high-intensity cardio workouts, offering a full-body exercise experience. Classes are structured to suit all skill levels and emphasize technique, strength, and flexibility. For those interested in competition, specialized training sessions are available to prepare members for kickboxing tournaments.

1.2 Objectives:

- The main objective of the project is to develop software that facilitates the data storage, data maintenance and its retrieval for the gym , boxing and kickboxing in an igneous way.
- To develop a user friendly system that requires minimal user training.
- Create detailed profiles for members and trainers, including personal information, fitness goals, diet and exercise.
- Allow trainers to manage and update their diet and exercise client interactions.
- To develop easy-to-use software which handles the customer-staff relationship in an effective manner.
- Also, only the staff has the privilege to access any database and make the required changes, if necessary.
- Develop an administrative dashboard for managing memberships, payments, and financial transactions.
- It also give an notification for renew membership , if it is expiring .

1.3 Purpose, Scope, and Applicability:

1.3.1 Purpose:

FitTrack Studio aims to be the ultimate online destination for all things related to fitness and gym activities. The purpose of FitTrack Studio is to provide users with comprehensive resources, tools, and services that cater to their fitness needs. Whether it's getting personalized diet plans, tracking fitness progress, or managing gym memberships, FitTrack Studio will offer a one-stop solution for fitness enthusiasts of all levels.

1.3.2 Scope:

1. **User Management:**
   - Develop profiles for members, trainers, and administrative staff.
   - Include features for registration, login, and profile management.
   - Implement role-based access control to ensure data privacy and security.

2. **Diet and Nutrition Management**:
   - Implement a feature for trainers to create, upload, and manage customized diet plans for their clients.
   - Enable members to access, download, and track their diet plans.

3. **Administrative Management:**
   - Design an administrative dashboard for managing memberships, payments, and financial transactions.
   - It also add the trainer and client information

4. **Maintenance and Support:**
   - Provide ongoing technical support and maintenance to ensure the system remains functional and up-to-date.
   - Implement a feedback mechanism for users to report issues and suggest improvement.

1.3.3 Applicability:

The Fittrack Studio Management System is highly applicable in enhancing the operational efficiency and member experience at Fittrack Studio. By offering personalized fitness and diet plans, it caters to the specific needs of each member, from beginners to advanced athletes. Trainers benefit from tools that help them create customized plans and monitor client progress, while administrative staff can efficiently manage resources and financial transactions. s. Overall, the Fittrack Studio Management System integrates all aspects of studio operations, promoting a seamless and effective fitness experience.

1.4 Achievements:

As an IT student taking on the role of a website maker for the Fittrack Studio, I have accomplished a range of achievements:

- Technical Proficiency: Leveraging my expertise as an IT student, I demonstrated strong technical skills in web development, programming languages, and database design, laying the foundation for the website's success.

- Problem-solving Abilities: Throughout the project, I applied my problem-solving techniques to overcome challenges, ensuring the website's functionality and user- friendliness.

- User Experience (UX) and User Interface (UI) Design: As both an IT student and website maker, I focused on delivering an intuitive and visually appealing website, enhancing the user experience for Fittrack Studio.

- Secure Data Management: I prioritized data security and privacy, implementing robust authentication mechanisms and ensuring compliance with data protection regulations.

- E-commerce Integration: As an IT student and website maker, I skillfully integrated a secure payment gateway, facilitating smooth and reliable financial transactions for Subscription.

In Conclusion my role as the website maker for Fittrack Studio allowed me to effectively apply and expand my IT knowledge and skills. I successfully developed a robust and dynamic website. I made an innate
and visually appealing platform that enhances user interaction. Secure data management was a top priority, ensuring the protection and privacy of user information. This comprehensive approach not only showcases my capabilities as an IT student but also underlining my commitment to delivering high-quality digital solutions.

1.5 Organisation of Report:

Chapter 1:

Introducing the structure of the thesis paper.

Chapter 2:

Explaining the technologies and tools, programming languages and methodology used in this project.

Chapter 3:

Explore the system requirement analysis (functional and non-functional requirements) and requirement specifications

Chapter 4:

Describing the system design of OSS and the topics are use-case diagrams, class diagram, sequence diagrams and database.

Chapter 5:

System implementation discusses the process of coding and implementation of the system.

Chapter 6:

System Testing discusses a practical example of electronic marking through the use on the Generic Interface System.

Chapter 7:-

Conclusion. recommendations and future work.

CHAPTER 2
SURVEY OF TECHNOLOGY

FRONT END / GUI TOOLS	.net Technologies, Java, HTML/CSS
DBMS / BACKEND	Oracle, SQL Plus, MySQL, SQL Server
Languages	C, C++, Java, C#, R, Python
Scripting Languages	PHP, JavaScript
.net Platform	C#, .net, Visual C# .net, ASP .net

1. **.net Technologies:**

   .NET is a free, cross-platform, open-source developer platform for building many different types of applications. With .NET, you can use multiple languages, editors, and libraries to build for web, mobile, desktop, games, and IoT. The first version of the .Net framework was 1.0 which came in the year 2002 .NET provides a standard set of base class libraries and APIs that are common to all .NET applications.

2. **Java:**

   Java is a high-level, object-oriented programming language that is widely used for a variety of applications, including web development, mobile apps, desktop software, enterprise systems, and more. It was originally developed by Sun Microsystems (acquired by Oracle Corporation) and released in 1995. Java's main features include platform independence, strong type checking, garbage collection, and a vast standard library.

3. **HTML/CSS:**

   HTML (Hyper Text Markup Language) and CSS (Cascading Style Sheets) are the fundamental building blocks of web development. HTML is used for creating the structure and content of a webpage, while CSS is used to style and format the elements of that webpage. Together, they provide the foundation for creating visually appealing

and functional websites.

DBMS/BACK END:

1. **Oracle:**
   Oracle Database is the first database designed for enterprise grid computing, the most flexible and cost-effective way to manage information and applications. Enterprise grid computing creates large pools of industry-standard, modular storage and servers. With this architecture, each new system can be rapidly provisioned from the pool of components. There is no need for peak workloads, because capacity can be easily added or reallocated from the resource pools as needed.

2. **MySQL:**
   MySQL is an Oracle-backed open-source relational database management system (RDBMS) based on Structured Query Language (SQL) that is widely used for storing and organizing data. It is known for its fast performance, reliability, and ease of use, making it a popular choice for web applications, business data, and other types of data storage.

3. **SQL Server:**
   SQL Server is a database server by Microsoft. It stands for Structured Query Language, a language for manipulating and talking about data in databases The Microsoft relational database management system is a software product which primarily stores and retrieves data as requested by other software applications on the same computer or a remote computer using the client-server model It works by carrying out different types of queries, or requests.

LANGUAGE:

1. **C:**
   C is a general-purpose programming language that is extremely popular, simple, and flexible to use. It is a structured programming language that is machine-independent and extensively used to write various applications, Operating Systems like Windows, and many other complex programs like Oracle database, Git, Python interpreter, and more.

2. **C++:**
   C++ is a high-level general-purpose programming language, created as

an extension of the original C language. It uses nearly identical syntax to C but adds new features for object- oriented programming language It also includes more advanced memory management features, like dynamic memory allocation, and a more comprehensive standard library of pre- written functions.

3. **C#:**

C# general-purpose, modern and object-oriented programming language pronounced as “C sharp”. C# is primarily used on the Windows .NET framework, although it can be applied to an open-source platform. It is a reliable crowd pleasure C# that can be used to create a number of different programs and applications: mobile apps, desktop apps, cloud-based services, websites, enterprise software and games. Lots and lots of games.

4. **Java:**

Java is a high-level, object-oriented programming language that is widely used for a variety of applications, including web development, mobile apps, desktop software, enterprise systems, and more. It was originally developed by Sun Microsystems (acquired by Oracle Corporation) and released in 1995. Java's main features include platform independence, strong type checking, garbage collection, and a vast standard library.

5. **Python:**

Python is an interpreted, object-oriented, high-level programming language with dynamic semantics Python is used for server-side web development, software development, mathematics, system scripting, data analysis, artificial intelligence, and scientific computing. Developers also use Python to build productivity tools, games, and desktop apps.

6. **Visual C++:**

Microsoft Visual C++ (MSVC) is the C++ programming language, which is part of Microsoft's Visual Studio Development environment, used to compile and develop Windows software. It was originally a standalone product but later became a part of Visual Studio and made available in both trialware and freeware forms. It features tools for developing and debugging C++ code, especially code written for the Windows API, DirectX, and .NET.

JUSTIFICATION OF PLATFORM:

Front End / GUI Tools	HTML, CSS, ASP .NET, C#
DBMS / Back End	MS-SQL
Languages	C#, Java
Scripting Languages	JavaScript, jQuery
Net Platform	ASP .Net

- HTML is a markup language used to create web pages and applications. When combined with JavaScript and CSS, HTML has become a milestone for web development. HTML helps to build the structure of a website and is a widely used Markup language. It is easy to learn, and every browser supports HTML Language. HTML is light weighted and fast to load. Storage of big files is allowed because of the application cache feature. You do not need to purchase any extra software because it’s by default in every window.
- Java is one of the most popular programming languages for web development. It is easy to handle and programmer's first choice for web development. Java enables developers to write once and use or reuse the code anywhere. This is one of the most important features and benefits of Java which makes it highly portable and thus it can be used in multi-platform applications easily[1]. Using Java as a programming language to develop web applications, you can ensure the security and the quality of the end product. Java offers numerous advantages for web developers such as its robustness and platform independence, widespread community support and numerous frameworks.
- C# is a general-purpose programming language used to build different types of applications and web-based services. It is easy to learn and simple to use. C# has a vast library that provides a higher level of functionality than other programming languages. C# is an object-oriented language which gives a clear structure to programs and allows code to be reused, lowering development costs. C# is also used for building .NET applications, which are common in Microsoft products.

- ASP.NET is a web application framework developed and marketed by Microsoft to allow programmers to build dynamic web sites, applications and services. It is built on the Common Language Runtime (CLR), allowing programmers to write ASP.NET code using any supported .NET language. ASP.NET is used to create web pages and web technologies and is an integral part of Microsoft's .NET framework vision. It provides a programming model, infrastructure, and tools for building enterprise-class web applications. ASP.NET is also used to build websites that are incredibly efficient, easily scalable and a breeze to maintain.

CHAPTER 3

SYSTEM ANALYSIS

3.1 Existing System

The previously existing systems faced numerous challenges that severely impacted efficiency and effectiveness. One major issue was the lack of security of data, as important information was often stored in unsecured or outdated formats, making it vulnerable to breaches, loss, or unauthorized access. Additionally, these systems required more manpower, with many tasks being carried out manually, from data entry to record-keeping, which not only increased operational costs but also heightened the chances of human error. This manual approach was also highly time-consuming, slowing down processes such as report generation, transaction handling, and everyday administrative tasks.

The reliance on physical documentation meant the system consumed a large volume of paperwork, which made storage, retrieval, and management cumbersome and inefficient. Many tasks, including financial tracking and data processing, required manual calculations, further increasing the likelihood of mistakes and making it difficult to maintain accuracy over time. Furthermore, there was no direct role for higher officials in overseeing or managing day-to-day operations, as the lack of digital integration prevented real-time monitoring and decision-making, leaving them detached from the core processes and unable to respond swiftly to emerging issues. These limitations underscored the need for a more automated, secure, and integrated approach to modernize operations.

3.2 Proposed System

The proposed system offers significant improvements by prioritizing the security of data, utilizing advanced encryption and secure storage methods to protect sensitive information from unauthorized access. It also ensures data accuracy by automating data input and validation, reducing errors that are common in manual processes. With proper control for higher officials, the system allows real-time monitoring and reporting, enabling them to make informed decisions quickly and effectively. By minimizing manual data entry, the system streamlines processes, ensuring that tasks are completed with minimum time needed for various processing, which leads to greater efficiency across operations.

Additionally, the system offers better service to both members and staff, as it speeds up transactions, bookings, and other interactions, enhancing overall satisfaction. The interface is designed to be user-friendly and interactive, making it easy for users to navigate the system and complete tasks without extensive training. Ultimately, the proposed solution focuses on minimizing delays, ensuring that minimum time is required for processing, thus improving both internal workflow and the member experience.

3.3 Requirement Analysis:

3.3.1 Problem Definition:

Before the introduction of the website, people faced several problems that hindered the efficiency and convenience of managing fitness facility operations. Data security was a major issue, with sensitive member and financial information often stored in unsecured or outdated formats, making it vulnerable to loss or unauthorized access. Manual data entry and processing were time-consuming, leading to frequent errors, delays, and inefficiencies in managing memberships, payments, and bookings. Additionally, the lack of real-time systems meant that higher officials had limited control over daily operations and decision-making, relying on fragmented or outdated reports.

The reliance on paperwork and manual calculations further slowed down processes, increasing the chances of mistakes and making record-keeping cumbersome. Members experienced poor service due to delays in bookings, payments, and scheduling, and the overall system lacked user-friendliness and interactivity, making it difficult for users to navigate and perform tasks easily. These issues underscored the need for a more streamlined, secure, and efficient system to improve operations and service delivery.

3.3.2 Requirement Specification:

The Requirement Specification explains the main features the new fitness facility management system should have to make it easy, secure, and efficient for users.

1 **User Management**:

- **Member Registration**: Allow users to register with their personal information (name, contact, etc.) and choose membership plans.
- **Trainer Registration**: Enable trainers to create profiles, including their qualifications, specializations, and availability.
- **Login/Authentication**: Secure login system for members, trainers, and admins using credentials (username, password, or social logins).
- **Role-Based Access**: Different access levels for admins, trainers, and members.

2 **Membership Management**:

- **Package Selection**: Members can choose from various membership packages and upgrade or renew their plans.
- **Membership Renewal Alerts**: Automatic notifications when a member's subscription is about to expire.
- **Membership Details**: View membership history, current plan, and attendance tracking.

3 **Booking and Scheduling**:

- **Session Booking**: Members can book gym sessions, personal training, boxing, and kickboxing classes based on trainer availability.

- **Class Schedules**: Trainers can set and update their schedules; members can view and book classes in real-time.
- **Event Registration**: Members can sign up for special events or workshops.

4 **Payment and Billing**:

- **Payment Gateway Integration**: Secure online payment options for membership fees, booking fees, or event participation.
- **Invoice Generation**: Automatic invoices for payments made, along with email receipts.
- **Subscription Management**: Members can view and manage their payment history and upcoming dues.

5 **Diet and Training Plans**:

- **Personalized Diet Chart**: Trainers can create and update customized diet plans for members.
- **Workout Plans**: Trainers can assign tailored workout plans to members based on their fitness goals and track progress.

6 **Feedback System**:

- **Feedback Form**: Members can submit feedback on trainers, classes, and services.
- **Feedback Analytics**: Admins and trainers can view feedback and ratings for improvement.

7 **Event Management**:

- **Event Creation**: Admins can create, schedule, and promote fitness events or challenges.
- **Event Registration**: Members can view upcoming events and register directly through the system.

8 **Admin Dashboard**:

- **Reports and Analytics**: Generate reports on member activities, trainer schedules, and revenue.
- **User Management**: Admins can manage users, edit profiles, deactivate accounts, and view system activity logs.

This system will make it easier for users to manage bookings, payments, and training, while also giving staff the tools to organize and improve services.

3.4 Hardware Requirements:

▪ Processor: Intel i3 or above

▪ Ram: At least 6GB

▪ Hard Disk: 20GB or more

3.5 Software Requirements:

▪ Programming Language: ASP.NET using C#

▪ Operating System: Windows 10 / 11

▪ Database: Microsoft SQL server

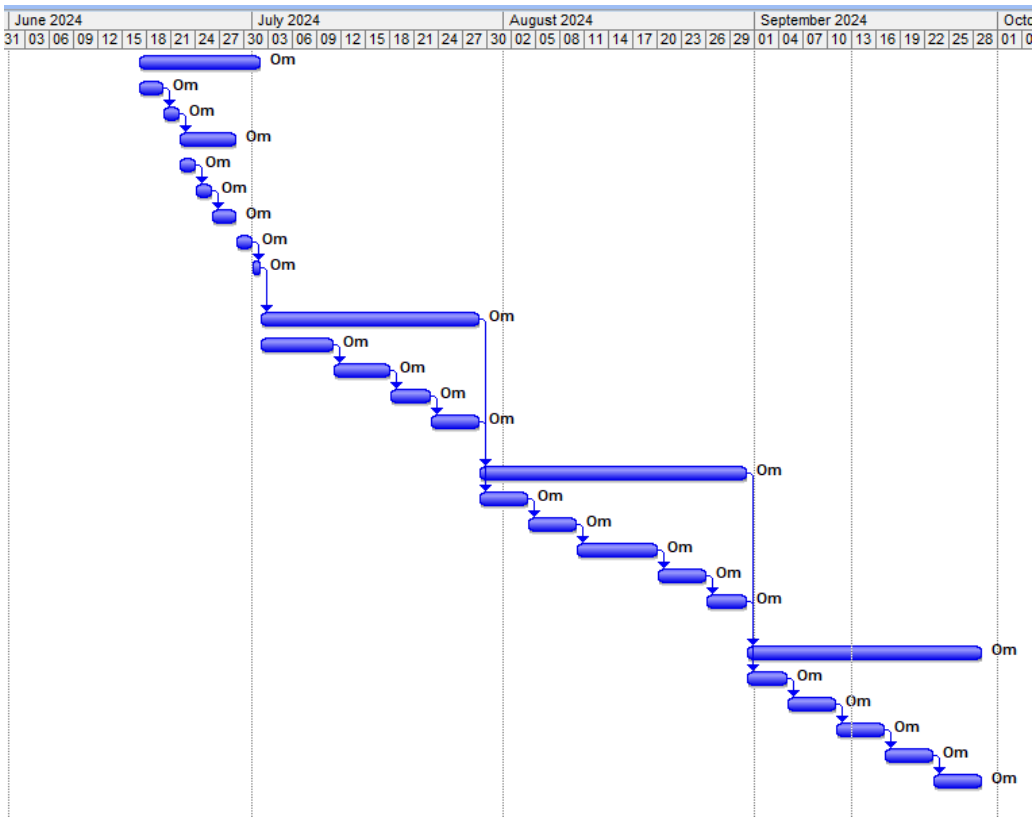
▪ Software: Visual Studio

▪ Browser Support: Firefox (latest) / Chrome (latest)

3.6 Planning & Scheduling:

Gantt Chart:

		Task Name	Duration	Start	Finish	Predecessors	Resource Names
1		Chapter 1 Introduction	15 days	Mon 17/06/24	Mon 01/07/24		Om
2		1.1 Background	3 days	Mon 17/06/24	Wed 19/06/24		Om
3		1.2 Objectives	2 days	Thu 20/06/24	Fri 21/06/24	2	Om
4		1.3 Purpose, Scope and A	7 days	Sat 22/06/24	Fri 28/06/24	3	Om
5		1.3.1 Purpose	2 days	Sat 22/06/24	Sun 23/06/24		Om
6		1.3.2 Scope	2 days	Mon 24/06/24	Tue 25/06/24	5	Om
7		1.3.3 Applicability	3 days	Wed 26/06/24	Fri 28/06/24	6	Om
8		1.4 Achivements	2 days	Sat 29/06/24	Sun 30/06/24		Om
9		1.5 Organisation of Report	1 day	Mon 01/07/24	Mon 01/07/24	8	Om
10							
11		Chapter 2 Survey of Techı	27 days	Tue 02/07/24	Sun 28/07/24	9	Om
12		2.1 Frontend / GUI	9 days	Tue 02/07/24	Wed 10/07/24		Om
13		2.2 Backend	7 days	Thu 11/07/24	Wed 17/07/24	12	Om
14		2.3 Languages	5 days	Thu 18/07/24	Mon 22/07/24	13	Om
15		2.4 Justification of Platforn	6 days	Tue 23/07/24	Sun 28/07/24	14	Om
16							
17		Chapter 3 System Analysis	33 days	Mon 29/07/24	Fri 30/08/24	15	Om
18		3.1 Existing System	6 days	Mon 29/07/24	Sat 03/08/24	11	Om
19		3.2 Proposed System	6 days	Sun 04/08/24	Fri 09/08/24	18	Om
20		3.3 Requirement Analysis	10 days	Sat 10/08/24	Mon 19/08/24	19	Om
21		3.4 Hardware Requiremen	6 days	Tue 20/08/24	Sun 25/08/24	20	Om
22		3.5 Software Requirement	5 days	Mon 26/08/24	Fri 30/08/24	21	Om
23							
24		Chapter 4 System Design	29 days	Sat 31/08/24	Sat 28/09/24	22	Om
25		4.1 Module Division	5 days	Sat 31/08/24	Wed 04/09/24	17	Om
26		4.2 Data Dictionary	6 days	Thu 05/09/24	Tue 10/09/24	25	Om
27		4.3 Entity Relatyionship Dia	6 days	Wed 11/09/24	Mon 16/09/24	26	Om
28		4.4 Data Flow Diagram	6 days	Tue 17/09/24	Sun 22/09/24	27	Om
29		4.5 UML Diagram	6 days	Mon 23/09/24	Sat 28/09/24	28	Om
30							
31							
32							
33							
34							
35							
36							
37			1 day?	Fri 09/08/24	Fri 09/08/24		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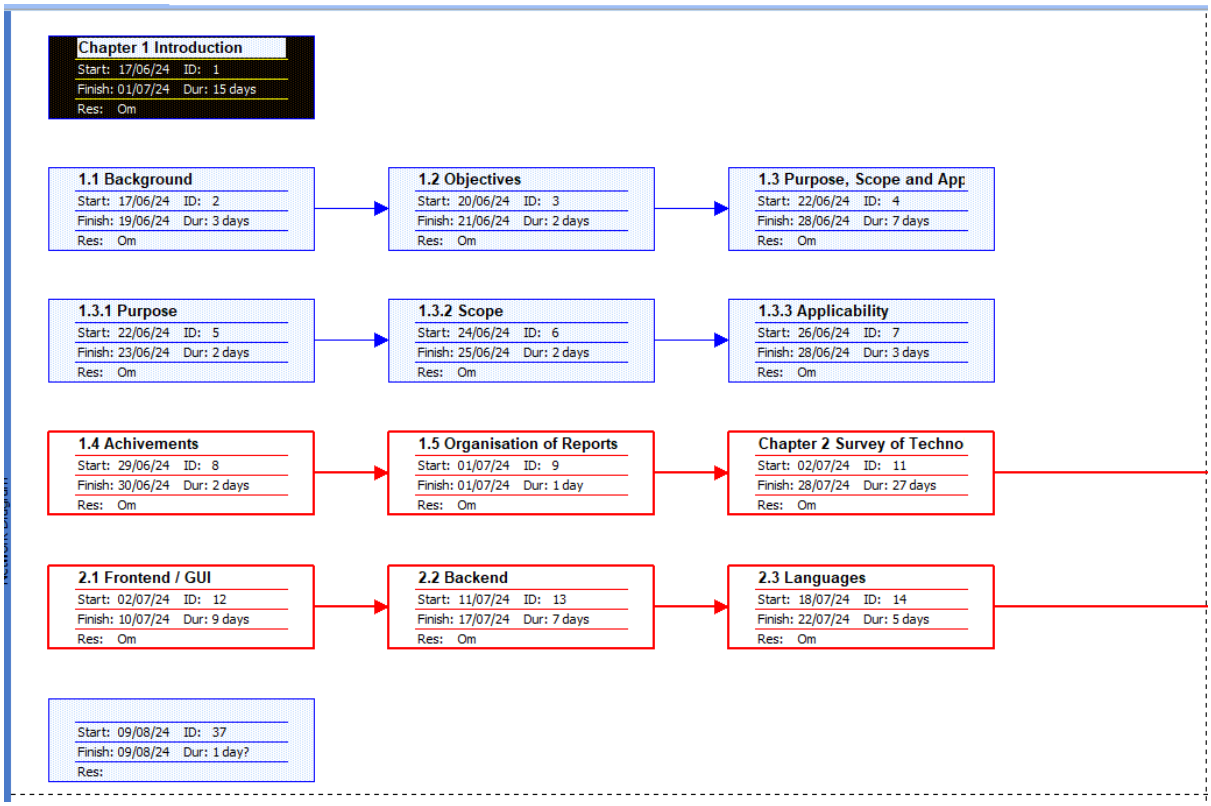
Om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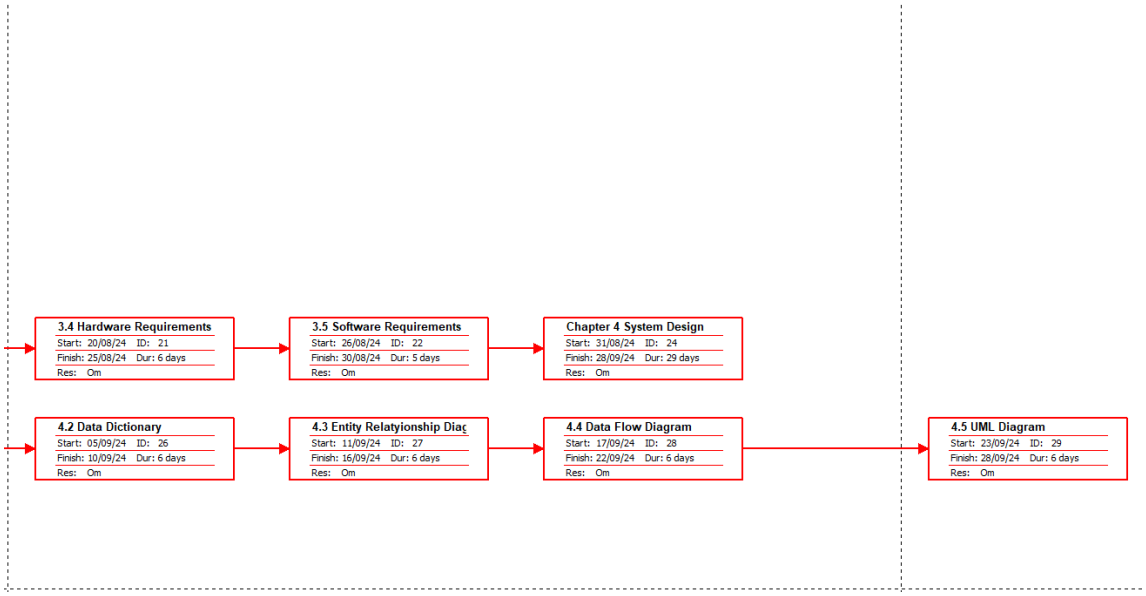
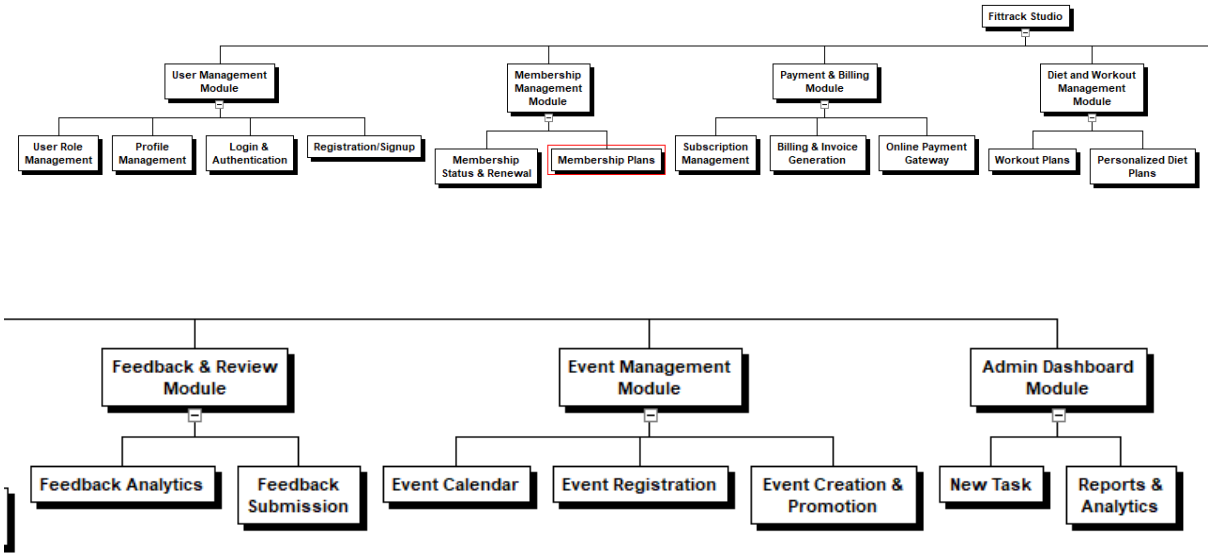
Res: Om
1.1 Background
Start: 17/06/24 ID: 2
Finish: 19/06/24 Dur: 3 days
Res: Om
1.2 Objectives
Start: 20/06/24 ID: 3
Finish: 21/06/24 Dur: 2 days
Res: Om
1.3 Purpose, Scope and App
Start: 22/06/24 ID: 4
Finish: 28/06/24 Dur: 7 days
Res: Om
1.3.1 Purpose
Start: 22/06/24 ID: 5
Finish: 23/06/24 Dur: 2 days
Res: Om
1.3.2 Scope
Start: 24/06/24 ID: 6
Finish: 25/06/24 Dur: 2 days
Res: Om
1.3.3 Applicability
Start: 26/06/24 ID: 7
Finish: 28/06/24 Dur: 3 days
Res: Om
1.4 Achivements
Start: 29/06/24 ID: 8
Finish: 30/06/24 Dur: 2 days
Res: Om
1.5 Organisation of Reports
Start: 01/07/24 ID: 9
Finish: 01/07/24 Dur: 1 day
Res: Om
Chapter 2 Survey of Techno
Start: 02/07/24 ID: 11
Finish: 28/07/24 Dur: 27 days
Res: Om
2.1 Frontend / GUI
Start: 02/07/24 ID: 12
Finish: 10/07/24 Dur: 9 days
Res: Om
2.2 Backend
Start: 11/07/24 ID: 13
Finish: 17/07/24 Dur: 7 days
Res: Om
2.3 Languages
Start: 18/07/24 ID: 14
Finish: 22/07/24 Dur: 5 days
Res: Om
Start: 09/08/24 ID: 37
Finish: 09/08/24 Dur: 1 day?
Res:


3.4 Hardware Requirements
Start: 20/08/24 ID: 21
Finish: 25/08/24 Dur: 6 days
Res: Om
3.5 Software Requirements
Start: 26/08/24 ID: 22
Finish: 30/08/24 Dur: 5 days
Res: Om
Chapter 4 System Design
Start: 31/08/24 ID: 24
Finish: 28/09/24 Dur: 29 days
Res: Om
4.2 Data Dictionary
Start: 05/09/24 ID: 26
Finish: 10/09/24 Dur: 6 days
Res: Om
4.3 Entity Relatyionship Diag
Start: 11/09/24 ID: 27
Finish: 16/09/24 Dur: 6 days
Res: Om
4.4 Data Flow Diagram
Start: 17/09/24 ID: 28
Finish: 22/09/24 Dur: 6 days
Res: Om
4.5 UML Diagram
Start: 23/09/24 ID: 29
Finish: 28/09/24 Dur: 6 days
Res: Om

Chapter 4

System Design

4.1 Module Division:



4.2 Data Dictionary:

1. Login Tabel

Column Name	Data Type	Description
login_id	INT	Primary Key, Auto-Increment
username	VARCHAR	User's unique username
password	VARCHAR	Encrypted password
user_role	VARCHAR	Role of user
last_login	DATETIME	Timestamp of the last login
member_id	INT	Foreign Key
trainer_id	INT	Foreign Key

2. Trainer Details Tabel

Column Name	Data Type	Description
Trainer_id	INT	Primary Key
First_name	VARCHAR	Trainer's first name
Last_name	VARCHAR	Trainer's last name
Specialization	VARCHAR	Trainer's expertise
Contact_info	VARCHAR	Contact Info
DateOfJoining	DATE	Date The Trainer Joined

3. Members Details Table

Column Name	Data Type	Description
member_id	INT	Primary Key
First_name	VARCHAR	Member's first name
Last_name	VARCHAR	Member's last name
Date_Of_Birth	DATE	Member's Date of birth
Contact_info	VARCHAR	Member's Contact Info
EmailId	VARCHAR	Member's email address
DateOfJoining	DATE	Date The Member Joined
Membership_Type	VARCHAR	Type of membership selected

4. Package Details Tabel

Column Name	Data Type	Description
Package_id	INT	Primary Key
package_name	VARCHAR	Name Of Packag
Duration_months	INT	Duration of Package
Price	DECIMAL	Price of package
Description	TEXT	Detailed Discription

5. Booking Details Tabel

Column Name	Data Type	Description
booking_id	INT	Primary Key
Member_id	INT	Foreign key
Trainer_id	INT	Foreign key
Package_id	INT	Foreign key
Booking_date	DATETIME	Date and Time of booking
Status	ENUM	Status of booking

6. Payment Details Tabel

Column Name	Data Type	Description
payment_id	INT	Primary Key
Booking_id	INT	Foreign key
Payment_date	DATE	Date when the payment made
Amount	DECIMAL	Total amount paid

7. Diet Chart Tabel

Column Name	Data Type	Description
diet_id	INT	Primary Key
member_id	INT	Foreign key
Trainer_id	INT	Foreign key
Diet_paln	TEXT	Detailed diet plan
Start_date	DATE	Starting date of diet
End_date	DATE	Ending date of diet

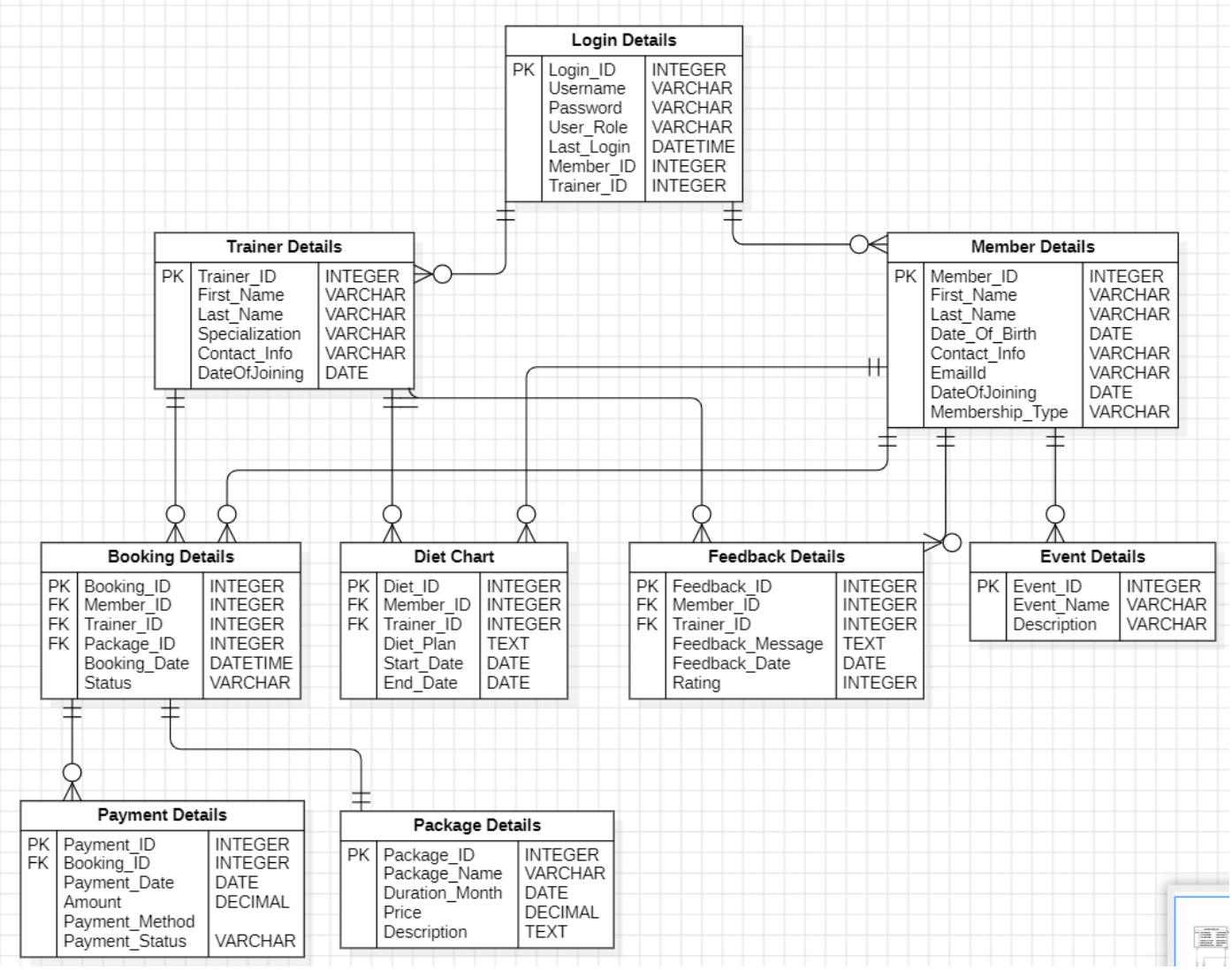
8. Feedback Details Tabel

Column Name	Data Type	Description
feedback_id	INT	Primary Key
member_id	INT	Foreign key
Trainer_id	INT	Foreign key
Feedback_Text	TEXT	Feedback gien by user,trainer
Feedback_date	DATE	The date when feedback is submitted

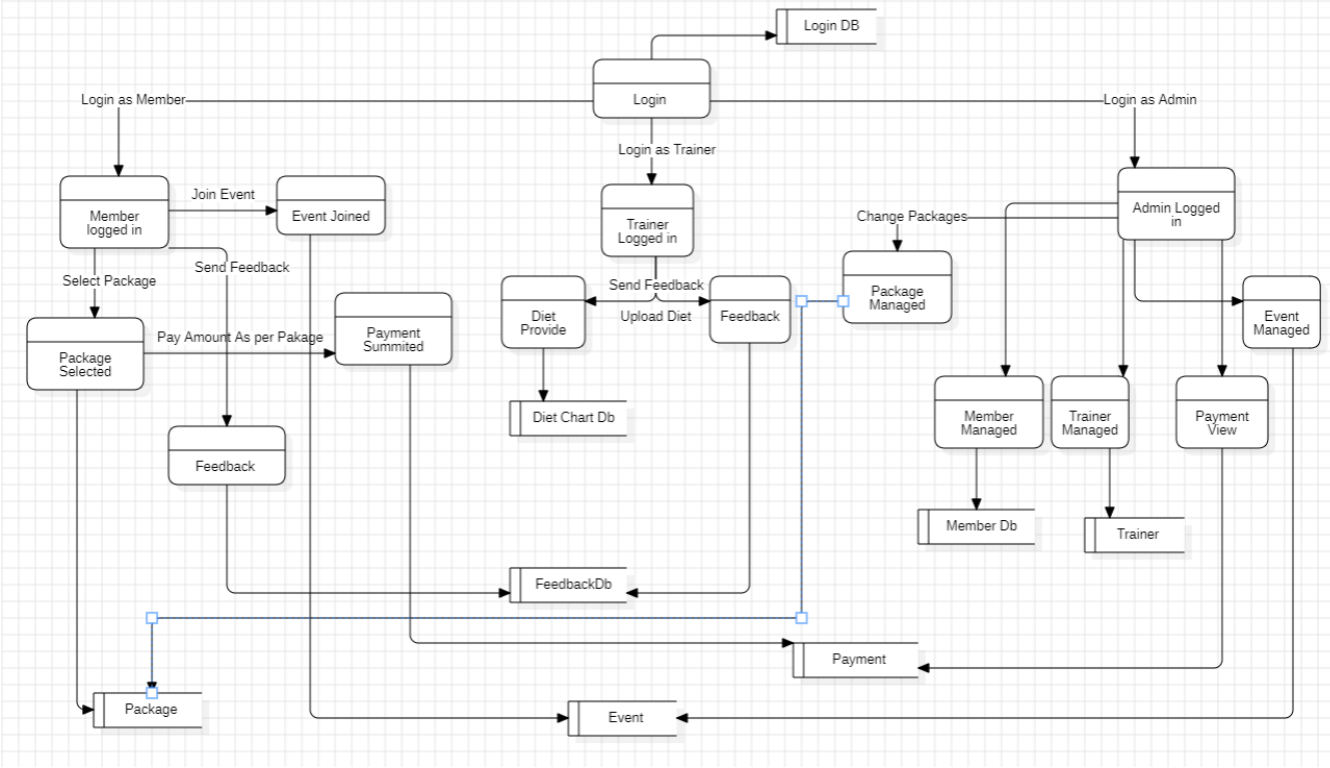
9. Event Detail Tabel

Column Name	Data Type	Description
event_id	INT	Primary Key
Event_name	VARCHAR	Name of event
Event_date	DATE	Date When event is scheduled
Description	TEXT	Description of event

4.3 Entity relationship diagram:



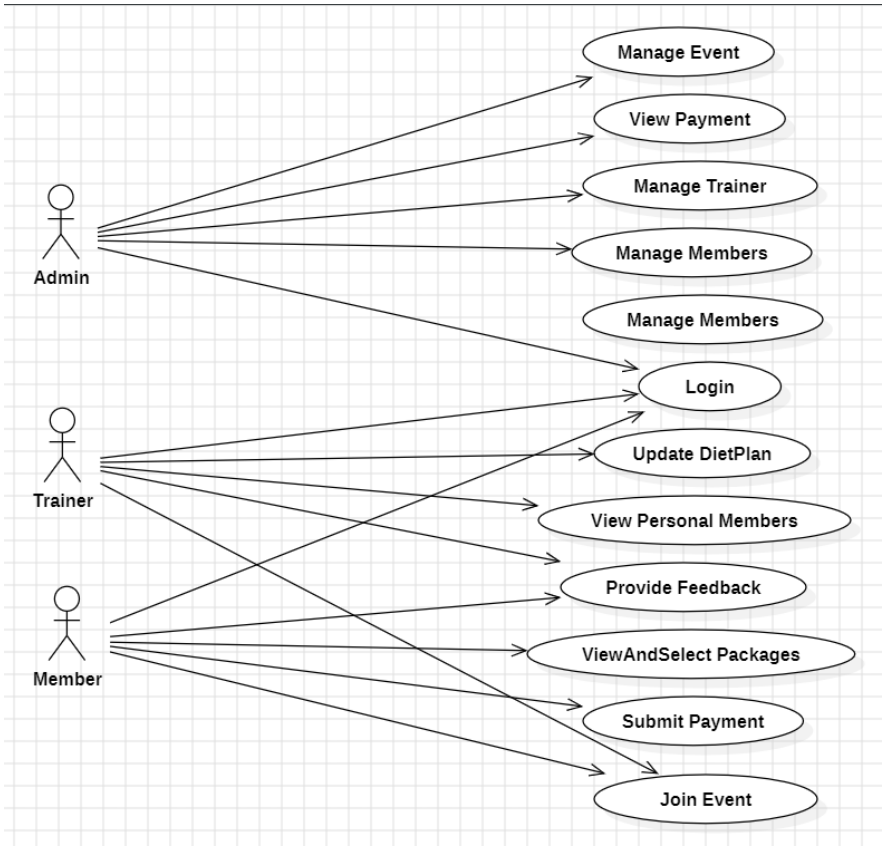
4.4 Data Flow diagram:



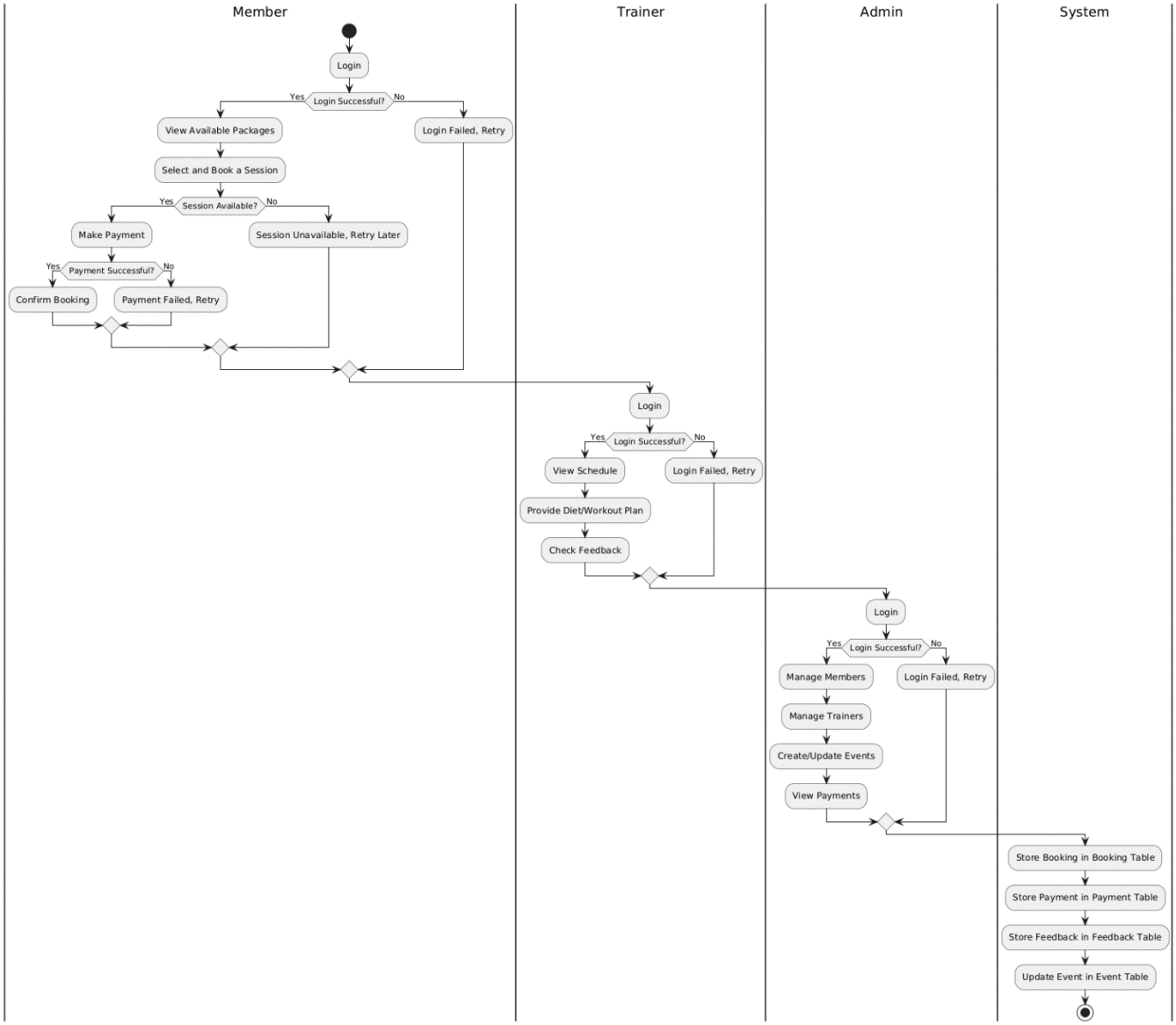
4.5 UML Diagrams

For Web Project and Mobile app

- **Use case diagram**

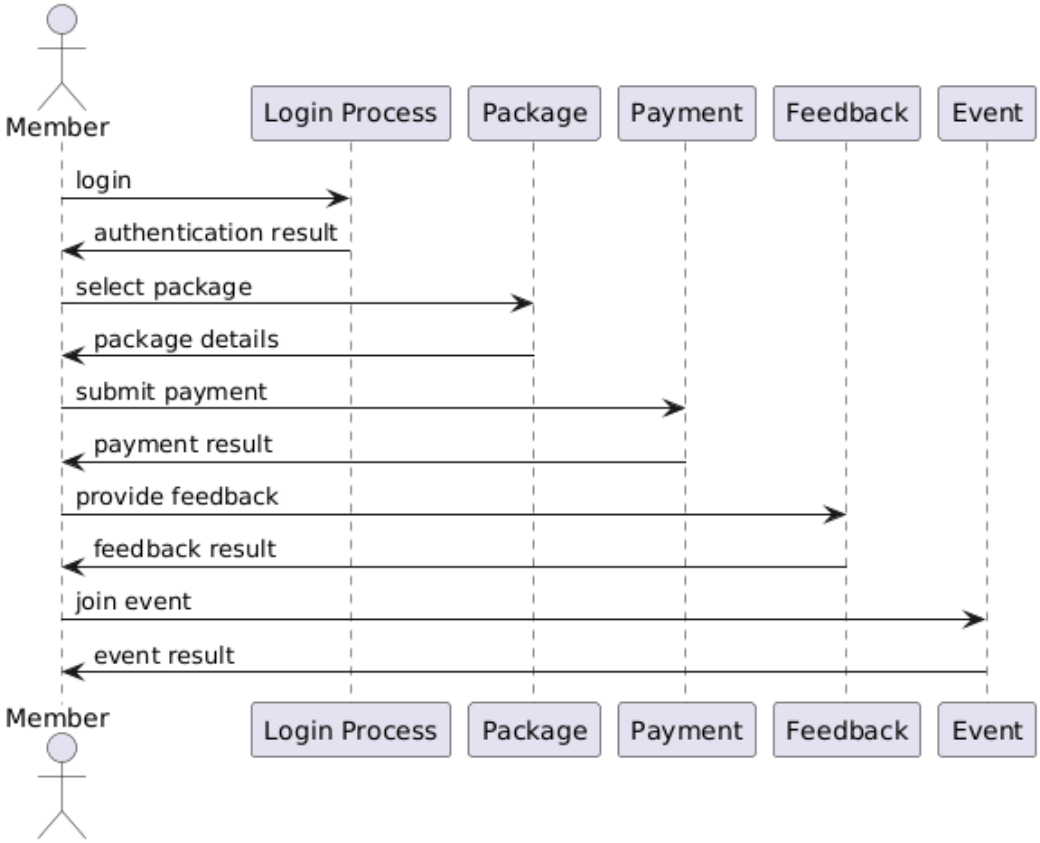


➢ Activity Diagram

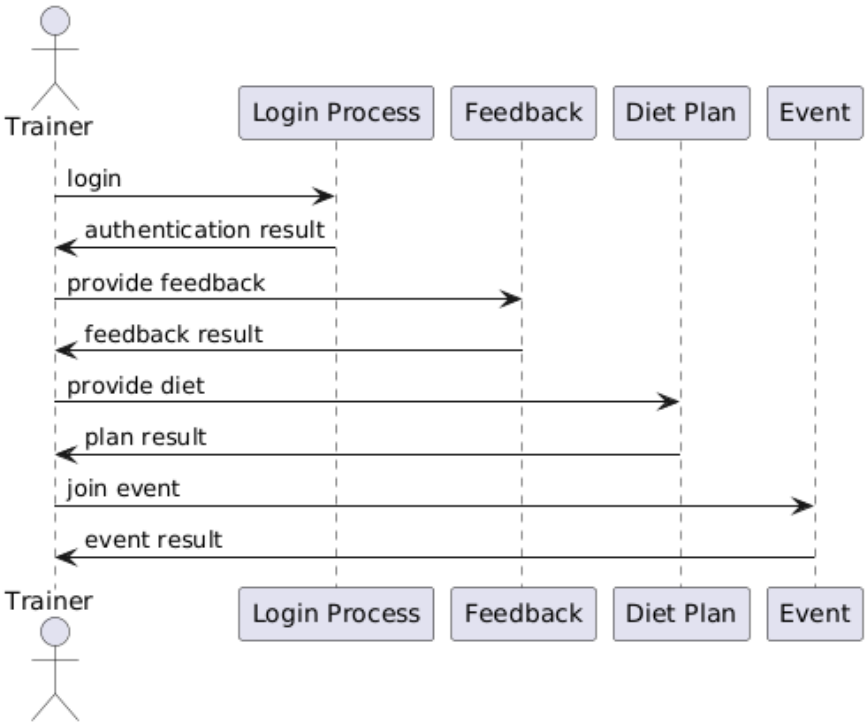


➢ **Sequence diagram**

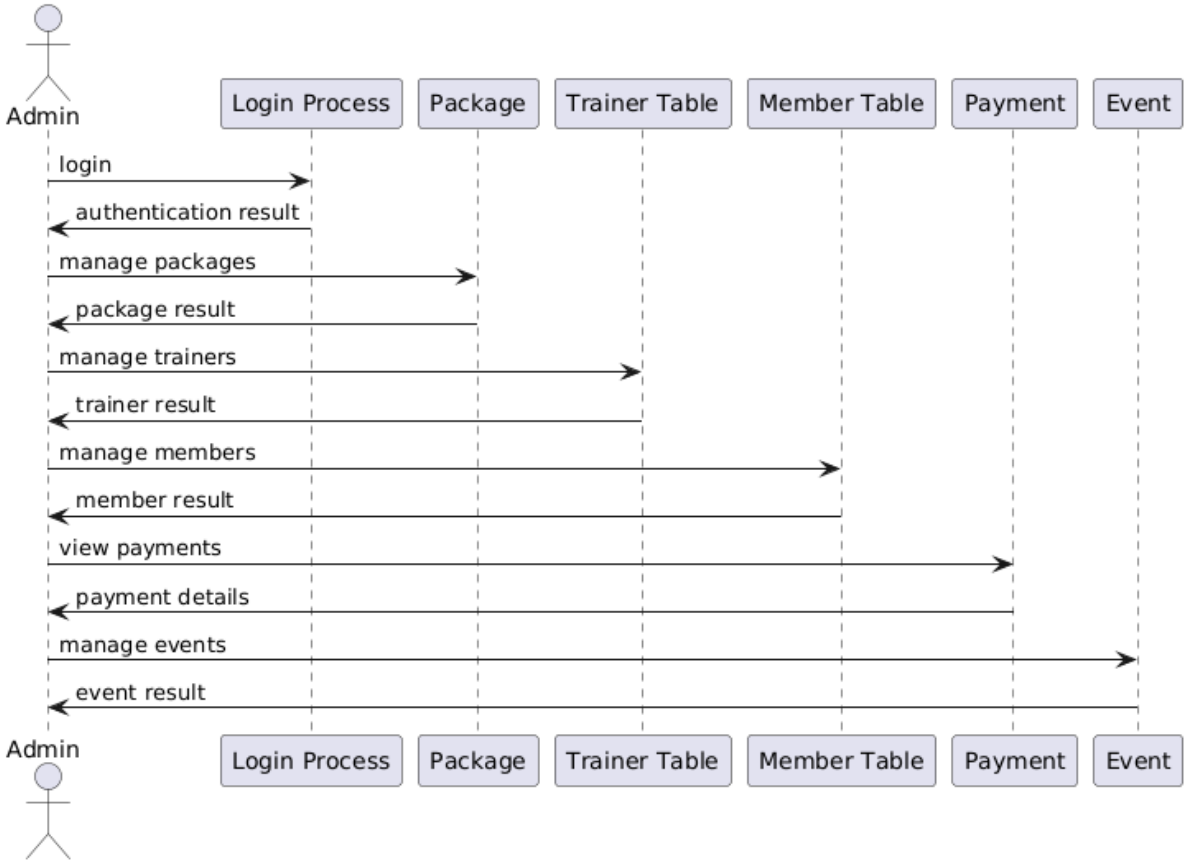
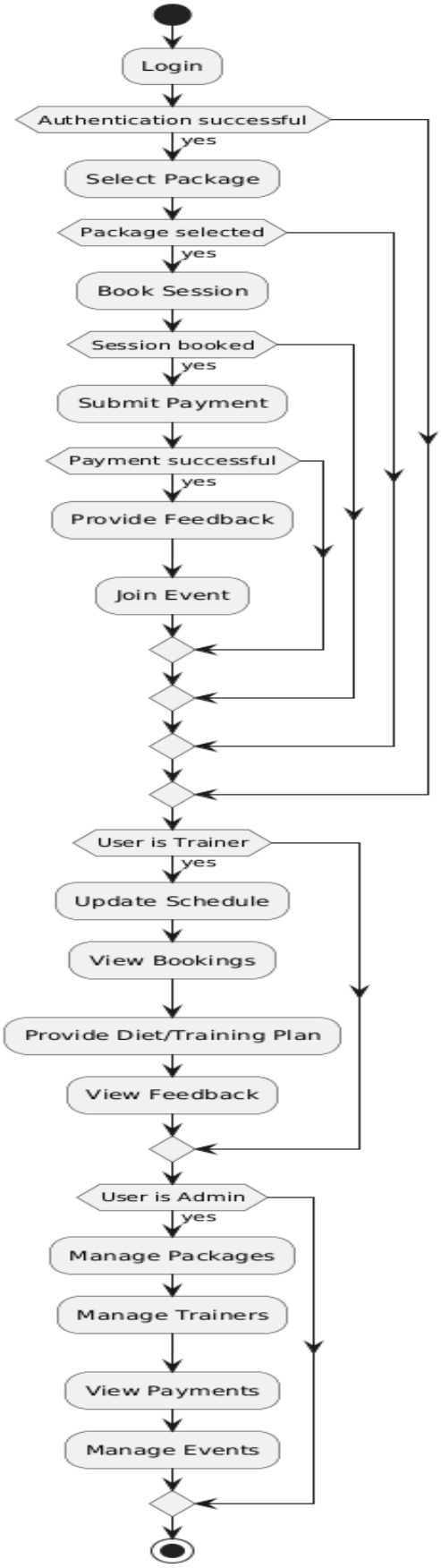
- **Member**



- **Trainer**



- **Admin**


Admin
Login Process
Package
Trainer Table
Member Table
Payment
Event
login
authentication result
manage packages
package result
manage trainers
trainer result
manage members
member result
view payments
payment details
manage events
event result
Admin
Login Process
Package
Trainer Table
Member Table
Payment
Event

➢ State Transition Diagram